Risk Analysis

Risk Exposures

- Sensitivity measures
  - Deviation of market prices due to a unit change of a specific market parameter
- Volatility measures
  - Variations around the average market price
- Downside measures of portfolio risk
  - Focuses only on the losses incurred by portfolios

Volatility Measure of Risk

$$\sigma(X) = \frac{\sqrt{\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2}}{n-1}; \overline{X} = \frac{\left(\sum_{i=1}^{n} X_i\right)}{n}$$

Note: Observed values = X ; Number of observations = n

Example

Day	Index	Return		Squared deviation from mean
	5846.03			
1	5662.6	-0.03138		0.00135790
2	5844.86	0.032187		0.00071363
3	5912.8	0.011624		0.00003784
4	5951.91	0.006614		0.00000130
5	6078.83	0.021324		0.00025127
6	6058.1	-0.00341		0.00007891
7	6209.76	0.025034		0.00038265
8	6146.15	-0.01024		0.00024701
9	6203.57	0.009342		0.00001497
10	6276.92	0.011824		0.00004033
11	6282.16	0.000835		0.00002151
12	6272.78	-0.00149		0.00004852
13	6191.25	-0.013		0.00034115
14	6269.95	0.012711		0.00005240
15	6333.39	0.010118		0.00002158
	Sum	0.082093	Sum	0.003611
	Average	0.005473	Variance	0.000258
			Standard deviation	0.01606

Volatility Measure of Risk

$$\sigma(X) = \sqrt{\sum_{i=1}^{n} p_i \left(X_i - \overline{X} \right)^2}, \overline{X} = \sum_{i=1}^{n} p_i X_i$$

Note: Observed values = X ; Number of observations = n; Probability of each X_i = p_i

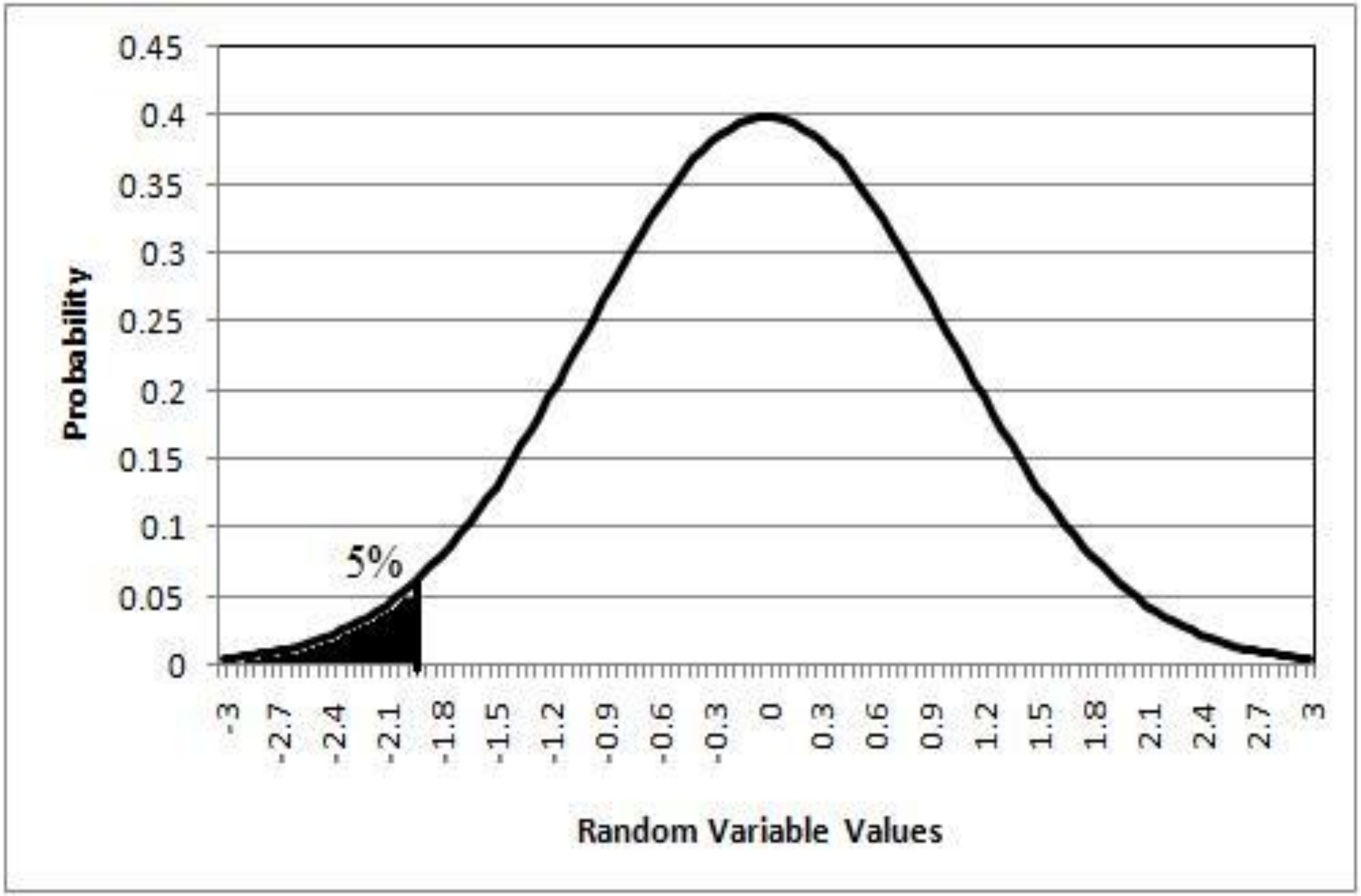
Example

Return	Probability	Return probability	Squared deviation from mean	
-0.03	0.02	-0.0006	0.001369	
-0.02	0.05	-0.001	0.000729	
-0.01	0.15	-0.0015	0.000289	
0.00	0.26	0	0.000049	
0.01	0.19	0.0019	0.000009	
0.02	0.17	0.0034	0.000169	
0.03	0.16	0.0048	0.000529	
	Total	Average	Variance	Standard deviation
	1	0.007	0.000235	0.01533

Value at Risk

- Probability of loss at a given level of confidence
- Downside measure of risk of an asset and considers only the left tail of the normal distribution

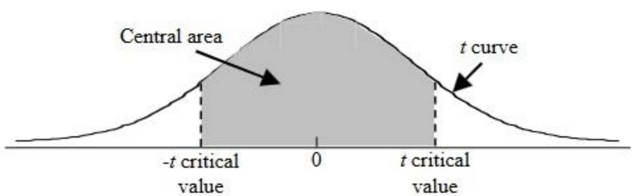
Example



0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
Probability
5%
-3
-2.7
-2.4
-2.1
-1.8
-1.5
-1.2
-0.9
-0.6
-0.3
0
0.3
0.6
0.9
1.2
1.5
1.8
2.1
2.4
2.7
3
Random Variable Values

Standard Normal Cumulative Distribution

t critical values



Confidence area captured: Confidence level:		0.90 90%	0.95 95%	0.98 98%	0.99 99%
Degrees of Freedom	1	6.31	12.71	31.82	63.66
	2	2.92	4.30	6.97	9.93
	3	2.35	3.18	4.54	5.84
	80	1.66	1.99	2.37	2.64
	90	1.66	1.99	2.37	2.63
	100	1.66	1.98	2.36	2.63
	1000	1.65	1.96	2.33	2.58
z critical values	∞	1.645	1.96	2.33	2.58
α for 2-tailed tests		0.10	0.05	0.02	0.01
α for 1-tailed tests		0.05	0.025	0.01	0.005

Example

- Parametric VaR
- Confidence interval 95%
- NORMINV[(1-0.95),0.051543, 0.098457]
- VaR=-0.1104; -11.04%

Year	Market Price	Dividend	Return
0	650	3	
1	690	3	0.066153846
2	710	3	0.033333333
3	695	3	-0.016901408
4	650	3	-0.060431655
5	720	3	0.112307692
6	825	4	0.151388889
7	706	4	-0.139393939
8	803	4	0.14305949
9	923	4	0.154420922
10	985	4	0.071505959

Average		0.051544313
Standard Deviation		0.098457068

Portfolio Risk and Return

$$\text{Portfolio return} = E(r) = \sum_{i=1}^{n} \omega_i r_i$$

$$\text{Portfolio risk} = \sqrt{\sum_{i=1}^{n} \omega_i^2 \sigma_{ii} + \sum_{i=1}^{n} \sum_{j=1}^{n} \omega_i \omega_j \sigma_{ij}}$$

where ω_i – Weight of a security; r_i – Security return;

n – Number of securities; σ_{ij} – Standard deviation

$$\sum_{i=1}^{n} \omega_i = 1$$

Computation of Portfolio Risk

Security	***1***	***2***	***3***	**********	***n***
1	$\omega_1\omega_1\sigma_{11}$	$\omega_1\omega_2\sigma_{12}$	$\omega_1\omega_3\sigma_{13}$	....	$\omega_1\omega_n\sigma_{1n}$
2	$\omega_2\omega_1\sigma_{21}$	$\omega_2\omega_2\sigma_{22}$	$\omega_2\omega_3\sigma_{23}$	....	$\omega_2\omega_n\sigma_{2n}$
3	$\omega_3\omega_1\sigma_{31}$	$\omega_3\omega_2\sigma_{32}$	$\omega_3\omega_3\sigma_{33}$	....	$\omega_3\omega_n\sigma_{3n}$
....	....	....	....	....	....
n	$\omega_n\omega_1\sigma_{n1}$	$\omega_n\omega_2\sigma_{n2}$	$\omega_n\omega_3\sigma_{n3}$	....	$\omega_n\omega_n\sigma_{nn}$

$$\sigma_p^2 = \boxed{\sum_{i=1}^{n} \omega_i^2 \sigma_{ii}} + \sum_{i=1}^{n}\sum_{j=1}^{n} \omega_i \ \omega_j \sigma_{ij}$$

$$\sigma_p = \sqrt{\sum_{i=1}^{n} \omega_i^2 \sigma_{ii} + \sum_{i=1}^{n}\sum_{j=1}^{n} \omega_i \omega_j \sigma_{ij}}$$

Example

Value and weights of securities

Security	1	2	3
Value	15000	15000	30000
Weight	0.25	0.25	0.5
Return	25%	18%	32%

Variance and covariance of securities

	1	2	3
1	0.45	0.24	0.65
2	0.24	0.36	0.31
3	0.65	0.31	0.52

Example

- Portfolio return

  (25% x 0.25)

  + (18% x 0.25)

  + (32% x 0.5)

  = 26.75%

Portfolio risk

	1	2	3	Variance	Standard deviation
1	0.028125	0.015	0.08125		
2	0.015	0.0225	0.03875		
3	0.08125	0.03875	0.13		
		Sum of diagonal	Sum of others		
		0.180625	0.27	0.450625	0.671286

Correlation

- $COV_{ij} = \frac{1}{N}\sum_{t=1}^{N}\left[(r_{it} - \overline{r_i}) \times \left(r_{jt} - \overline{r_j}\right)\right]$
- $COV_{ij} = \sigma_i \sigma_j \rho_{ij}$
- $\rho_{ij} = \frac{COV_{ij}}{\sigma_i \sigma_j}$
- $\rho_{ij} = \frac{N\sum XY - \sum X \sum Y}{\{[(N\sum X^2) - (\sum X)^2] \times [(N\sum Y^2) - (\sum Y)^2]\}^{0.5}}$

Example

- Correlation = 2750 / 2929
- Correlation = 0.9388958

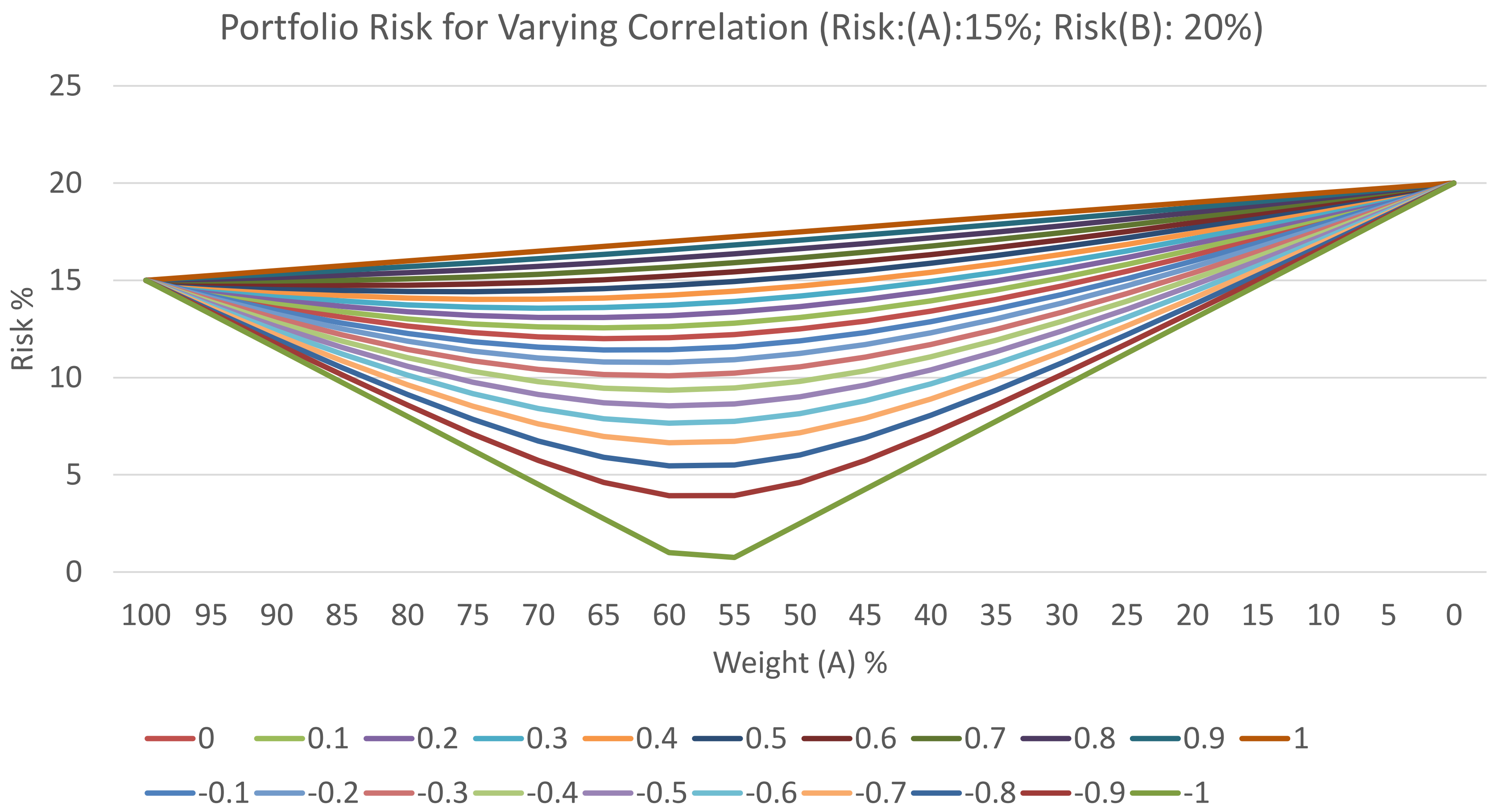
	Return X	Return Y	X^2	Y^2	XY
1	8	10	64	100	80
2	-9	-12	81	144	108
3	14	18	196	324	252
4	16	20	256	400	320
5	20	14	400	196	280
Sum	49	50	997	1164	1040
Correlation					0.938896

Example

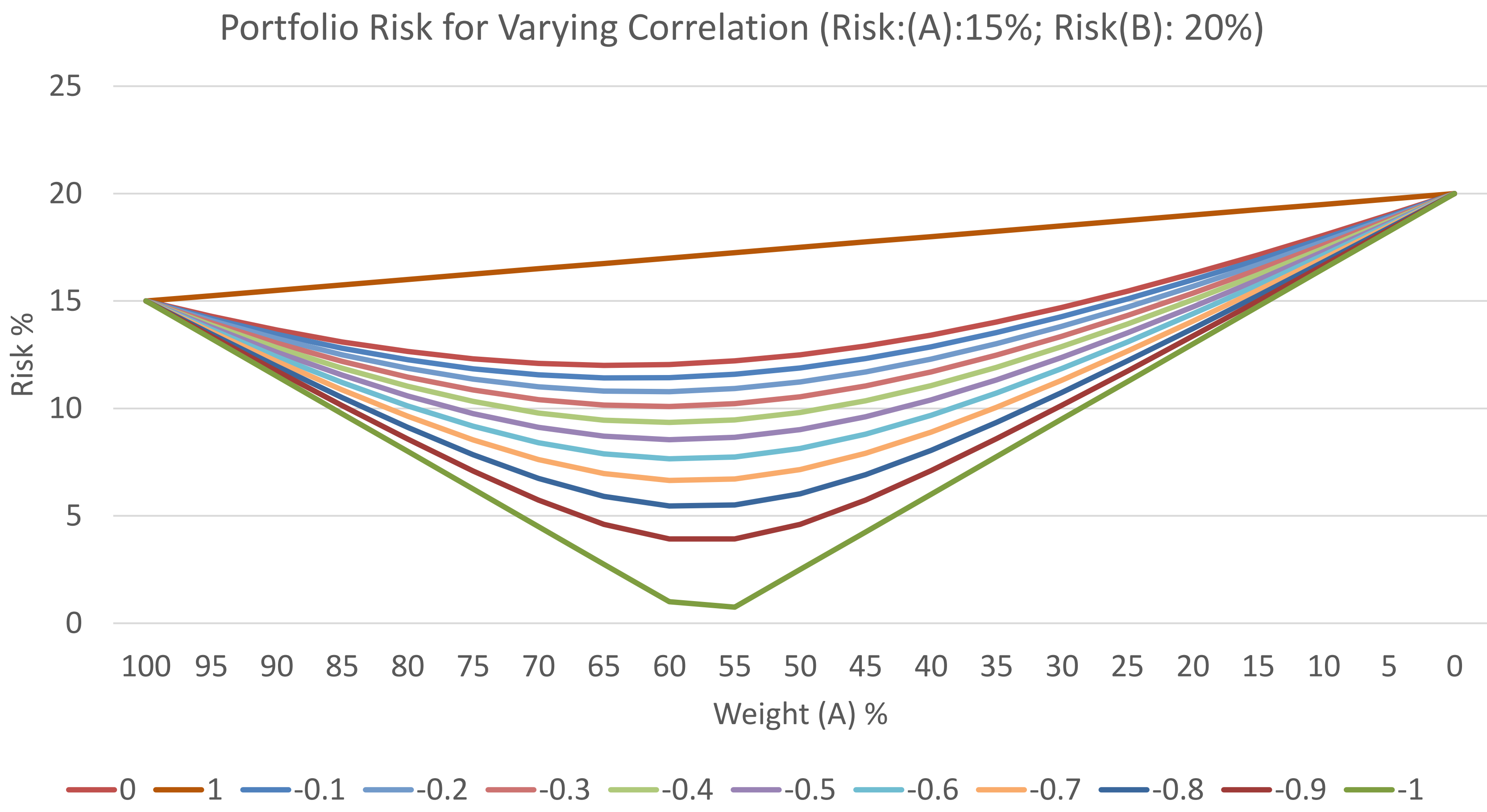
- Correlation = -465 / 482
- Correlation = -0.96474

	Return X	Return Y	X^2	Y^2	XY
1	5	20	25	400	100
2	8	16	64	256	128
3	11	11	121	121	121
4	15	10	225	100	150
5	17	8	289	64	136
Sum	56	65	724	941	635
Correlation					-0.96474

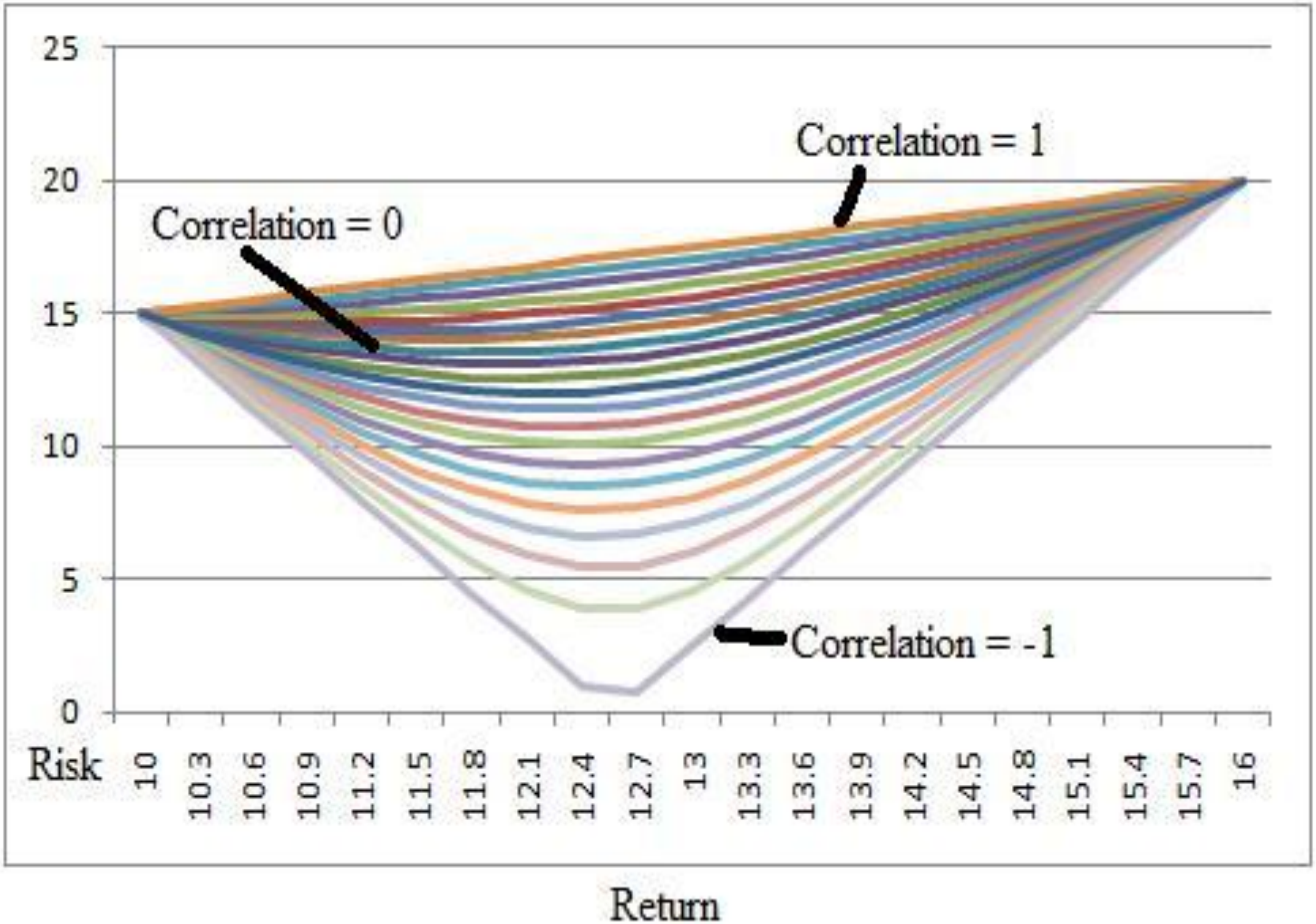
Positive and Negative Correlation Impact



Negative Correlation Impact



Possible risk-return characteristics of portfolios (Return (A): 10%; (B):16%)



Portfolio Risk::Return

- Compute risk–return characteristic of an equally weighted portfolio of three securities whose individual risk and return are:

Security	Risk (%)	Return (%)
A	15	12
B	20	18
C	25	22

- Correlation between security A and B is –0.43; correlation between security B and C is 0.21 and correlation coefficient between security A and C is –0.62

Portfolio Risk

- Portfolio return
- (0.33 x 12) + (0.33 x 18) + (0.33 x 22) = 17.16%
- Portfolio Risk

$$\sigma = \sqrt{\begin{array}{l}(0.33^2 \times 15^2) + (0.33^2 \times 20^2) + (0.33^2 \times 25^2) + (2 \times 0.33^2 \times -0.43 \times 15 \times 20) + \\ (2 \times 0.33^2 \times 0.21 \times 20 \times 25) + (2 \times 0.33^2 \times -0.62 \times 15 \times 25)\end{array}}$$

- = 8.96%

Minimum Risk Portfolio

- From the two securities available for investment opportunity, find the proportion of investment in each security that will minimize risk for the investor.

Security	Risk (%)	Return (%)
A	25	18
B	30	22

- Correlation coefficient between the two securities is –0.65.

Portfolio Risk::Return

- Min variance Portfolio

$$\omega_A = \frac{\sigma_B^2 - \rho_{AB}\sigma_A\sigma_B}{\sigma_A^2 + \sigma_B^2 - 2\rho_{AB}\sigma_A\sigma_B}$$

- Weight (A) = 0.55
- Weight (B) = 0.45
- Portfolio Return = 19.78%
- Portfolio Risk = 11.40%